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Oil Trade Flow Tracker

Real-Time AIS Data Pipeline, Geospatial Analytics, and Oil-on-Water Estimation

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GitHub Repository: <https://github.com/AlexisGpro/oil-trade-flow-tracker>

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Abstract

This report outlines the design and technical implementation of the **Oil Trade Flow Tracker**, a decision-support analytical tool engineered to monitor the global crude and product tanker fleet and estimate active volumes of *Oil-on-Water*. By leveraging 2D computational geometry (*Point-in-Polygon containment tests*) and naval hydrodynamics principles (*Draft-to-Max-Draft ratios and Archimedes' buoyancy*), the system detects maritime congestion at strategic chokepoints (e.g., Suez Canal, Strait of Hormuz) and delivers real-time market intelligence via an interactive Streamlit and Folium dashboard.

1 Introduction & Business Context

In energy trading and chartering desks (e.g., TotalEnergies / BxT Trading), anticipating freight rate volatility and crude price spreads depends heavily on tracking physical oil supply in transit. Telemetry signals broadcasted by the **AIS** (*Automatic Identification System*) provide real-time spatial coordinates, vessel operational status, and draft metrics across the global commercial fleet.

Key objectives of this project include:

1. Dynamic ingestion and spatial filtering of AIS telemetry data across tanker classes (VLCC, Suezmax, Aframax).
2. Automated detection of vessel transits through critical strategic passages known as **Chokepoints**.
3. Quantitative estimation of active crude oil volumes at sea (*Oil-on-Water*) calculated via immersion ratios.

2 Mathematical & Physical Modeling

2.1 Chokepoint Spatial Containment (Computational Geometry)

Each strategic chokepoint k is defined as a closed 2D polygon $P_k \subset \mathbb{R}^2$ constructed from N boundary GPS coordinates (longitude x , latitude y). A vessel i represented by its instantaneous position $M_i(x_i, y_i)$ is evaluated using a point-in-polygon spatial containment test:

$$\text{Chokepoint}(M_i) = \begin{cases} k & \text{if } M_i(x_i, y_i) \in P_k \\ \text{"Open Ocean"} & \text{otherwise} \end{cases} \quad (1)$$

2.2 Oil-on-Water Volume Estimation

The total volume of oil at sea V_{total} , expressed in Million Barrels (MMbbl), is derived from the ratio between real-time operational draft D_{real} and maximum design draft D_{max} :

$$\alpha_i = \frac{D_{\text{real},i}}{D_{\text{max},i}} \quad \text{where } 0 \leq \alpha_i \leq 1 \quad (2)$$

Applying the maximum carrying capacity (DWT_i) and the standard volumetric conversion factor for crude oil (1 metric ton $\approx$ 7.33 barrels):

$$V_{\text{total}} = \frac{7.33}{10^6} \times \sum_{i \in \text{Underway}} (\text{DWT}_i \times \alpha_i) \quad (3)$$

3 Software Architecture & Implementation

The application enforces a modular architecture to maintain clean separation of concerns:

- `src/geospatial_analytics.py`: Core computational geometry and physics engine.
- `app.py`: Decision-support web dashboard built using `Streamlit` and `Folium`.

3.1 Geospatial Engine Extract (`geospatial_analytics.py`)

```

1 import pandas as pd
2 from shapely.geometry import Point, Polygon
3
4 CHOKEPOINTS = {
5     "Suez_Canal": Polygon([(31.0, 27.0), (35.0, 27.0), (35.0, 32.0),
6     (31.0, 32.0)]),
7     "Strait_of_Hormuz": Polygon([(53.0, 24.0), (58.0, 24.0), (58.0,
8     27.5), (53.0, 27.5)])
9 }
10
11 def detect_chokepoints(df_ais: pd.DataFrame) -> pd.DataFrame:
12     chokepoint_list = []
13     for _, row in df_ais.iterrows():
14         point = Point(row['longitude'], row['latitude'])
15         found = "Open Ocean"
16         for name, polygon in CHOKEPOINTS.items():
17             if polygon.contains(point):
18                 found = name
19                 break
20         chokepoint_list.append(found)
21     df_ais['chokepoint'] = chokepoint_list
22     return df_ais

```

Listing 1: Chokepoint detection and volume estimation algorithm

4 Dashboard Results & Visual Analytics

The resulting web dashboard provides market analysts with:

- Top-level KPIs (*Tracked Tankers*, *Oil-on-Water*, *Congestion / Waiting*).
- An interactive real-time map built with `Folium` highlighting moored/waiting tankers in red and vessels underway in blue.

- Regional fleet breakdown visualizations generated via dynamic Plotly charts.

5 Conclusion & Future Work

This project demonstrates the capability to ingest, structure, and transform spatial AIS signals into actionable financial and operational intelligence for energy trading desks. Future developments will incorporate streaming ingestion via Apache Kafka and predictive arrival time (ETA) models for key discharge terminals.